Control statement-

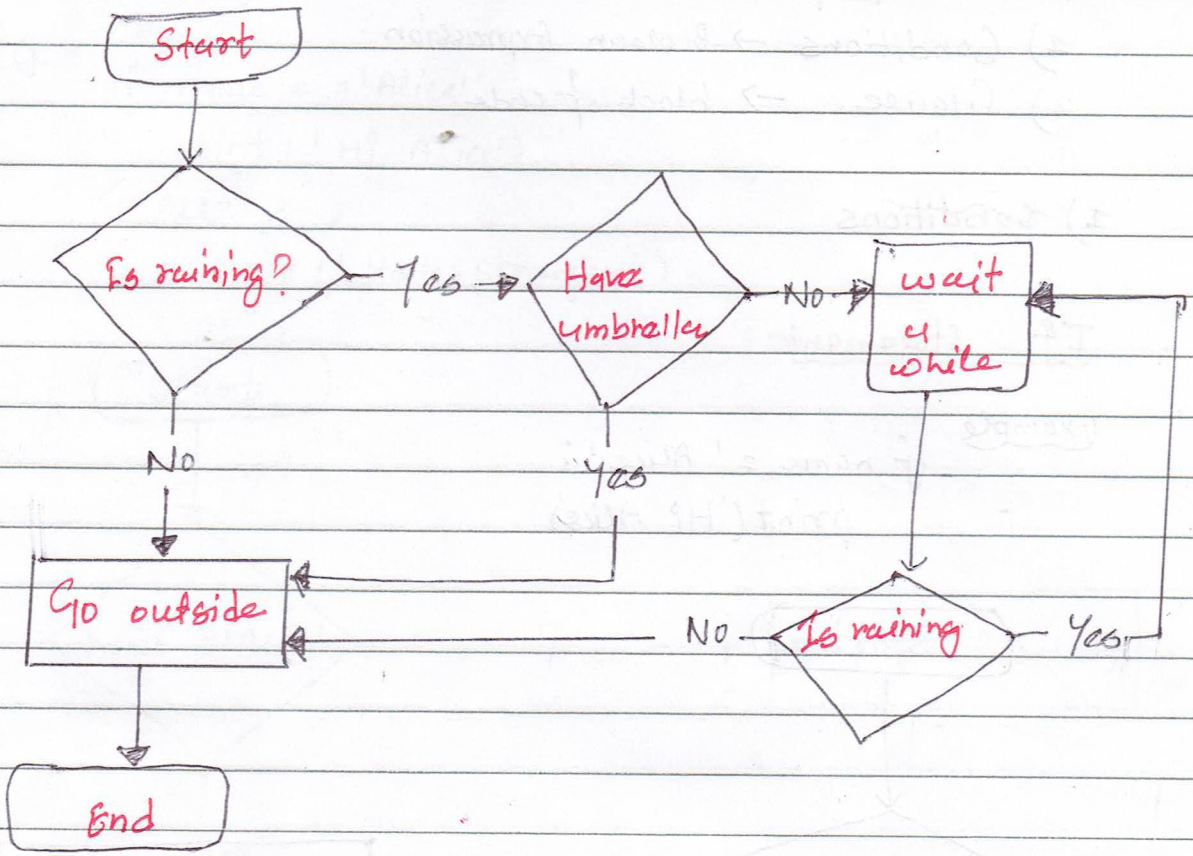


Fig: 2-1 : A flowchart to tell you what to do if it is raining.

→ Starting and Ending steps are represented with rounded rectangles.

→ Branching point is represented with diamond shape.

→ other steps are represented in rectange shape.

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Elements of flow control.

1) Conditions → Boolean Expression.
2) Clause → block of code.

1) Conditions

If Statement.

Example

```
if name = 'Alice':
    print('Hi Alice')
```

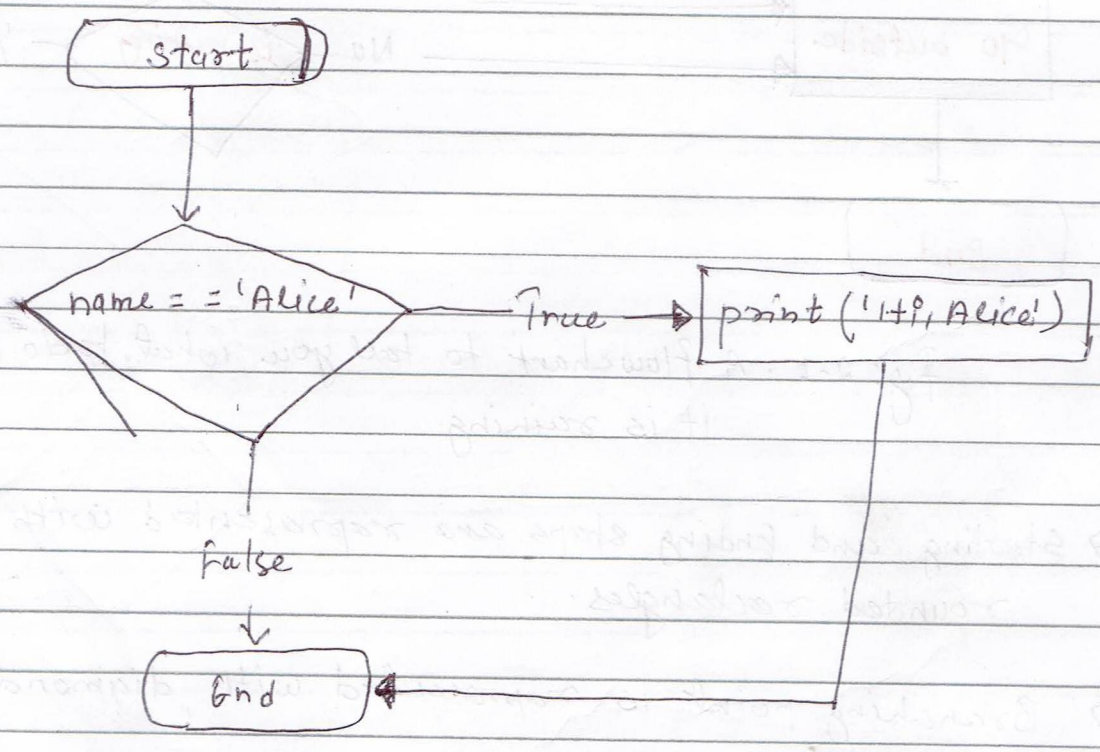


Fig: The flowchart for an if statement.

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Else - Statement

Eg:

```
if name == 'Alice':
    print('Hi, Alice!')
Else:
    print('Hallo, stranger').
```

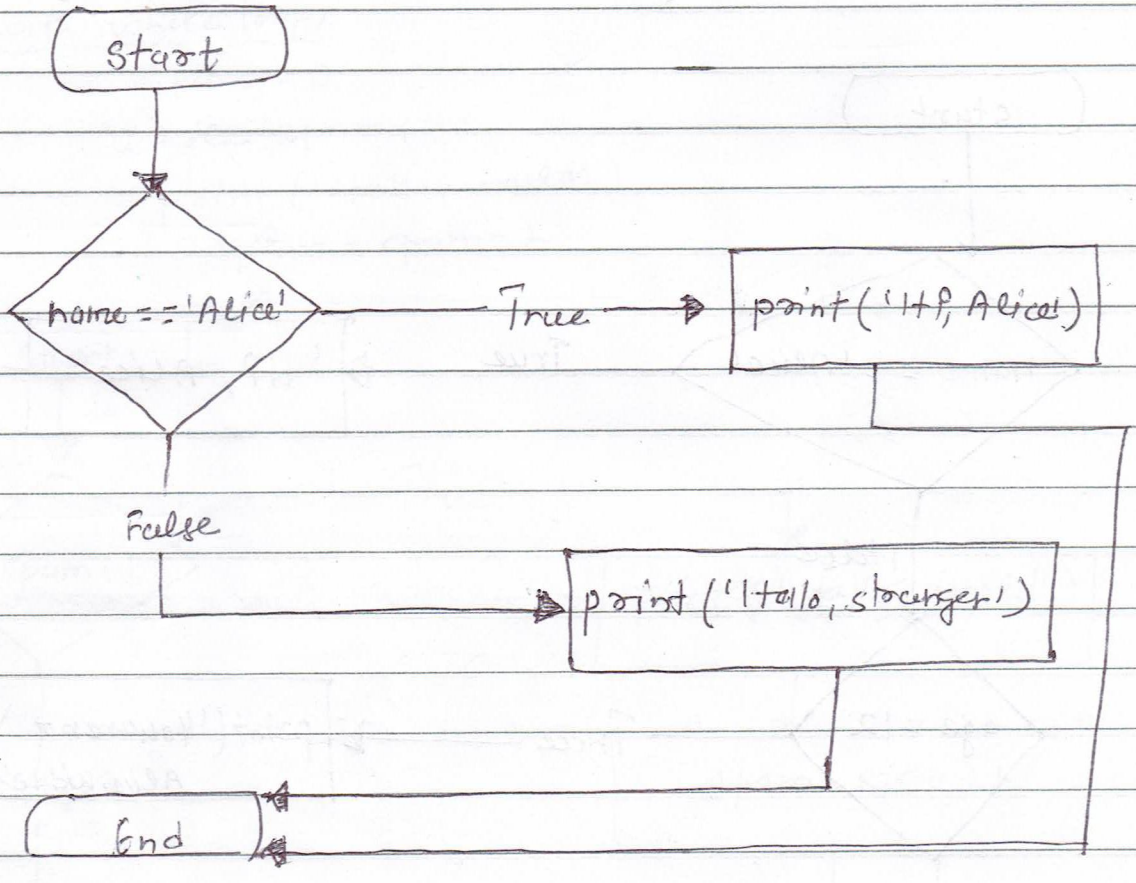


Fig: The flowchart for an else statement.

Fig: The flowchart for an else statement.

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Elseif Statement

E.g.
```
if name == 'Alice':
    print('Hi, Alice')
elif age < 12:
    print ('You are not Alice, kiddo.')
```

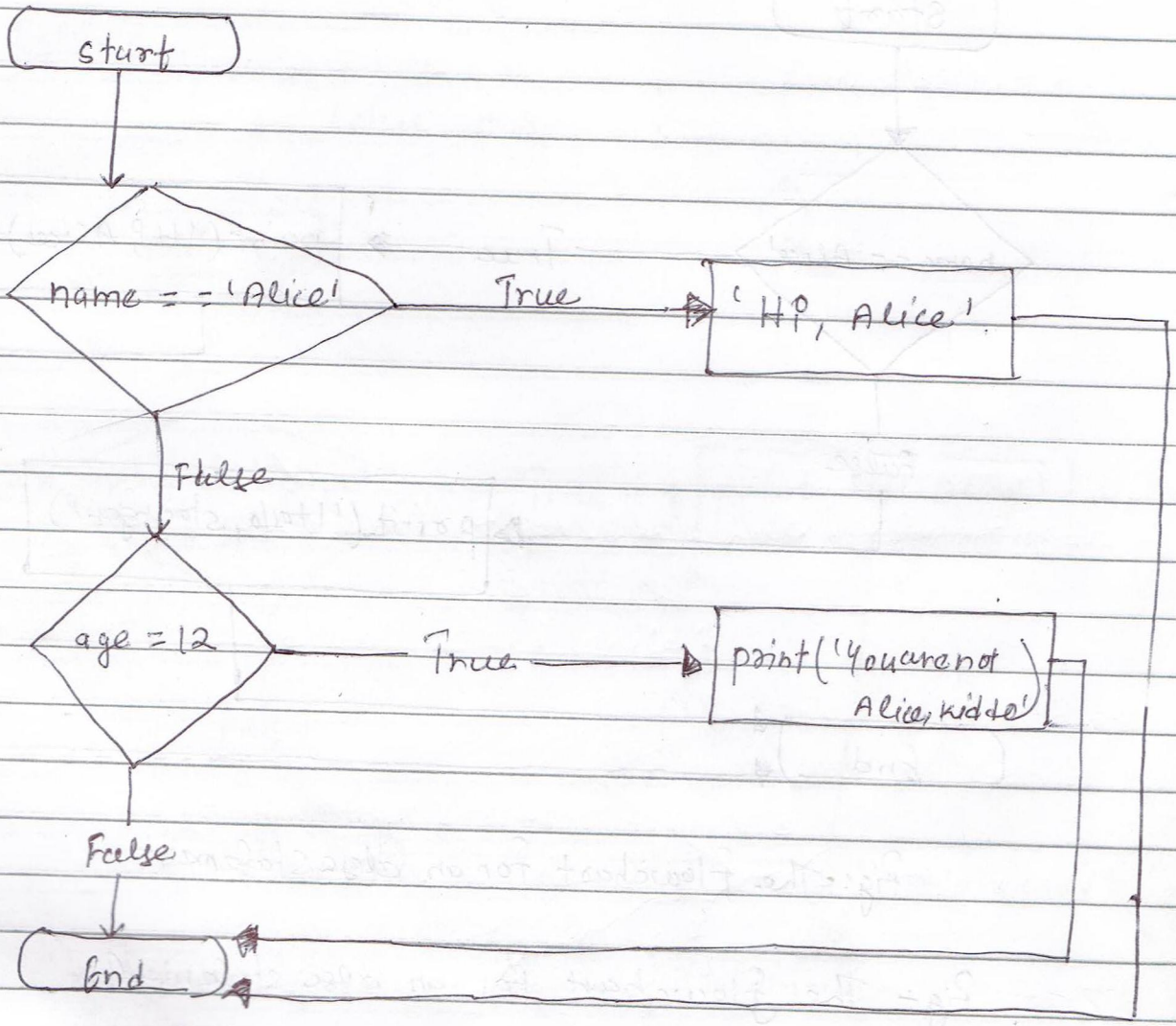


Fig: The Flowchart for an elif statement

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While loop statement.

E.g.

```
spam = 0
if spam < 5:
    print('Hello world')
    spam = spam + 1.
```

With while loop;

```
spam = 0
while spam < 5:
    print('Hello, world')
    spam = spam + 1.
```

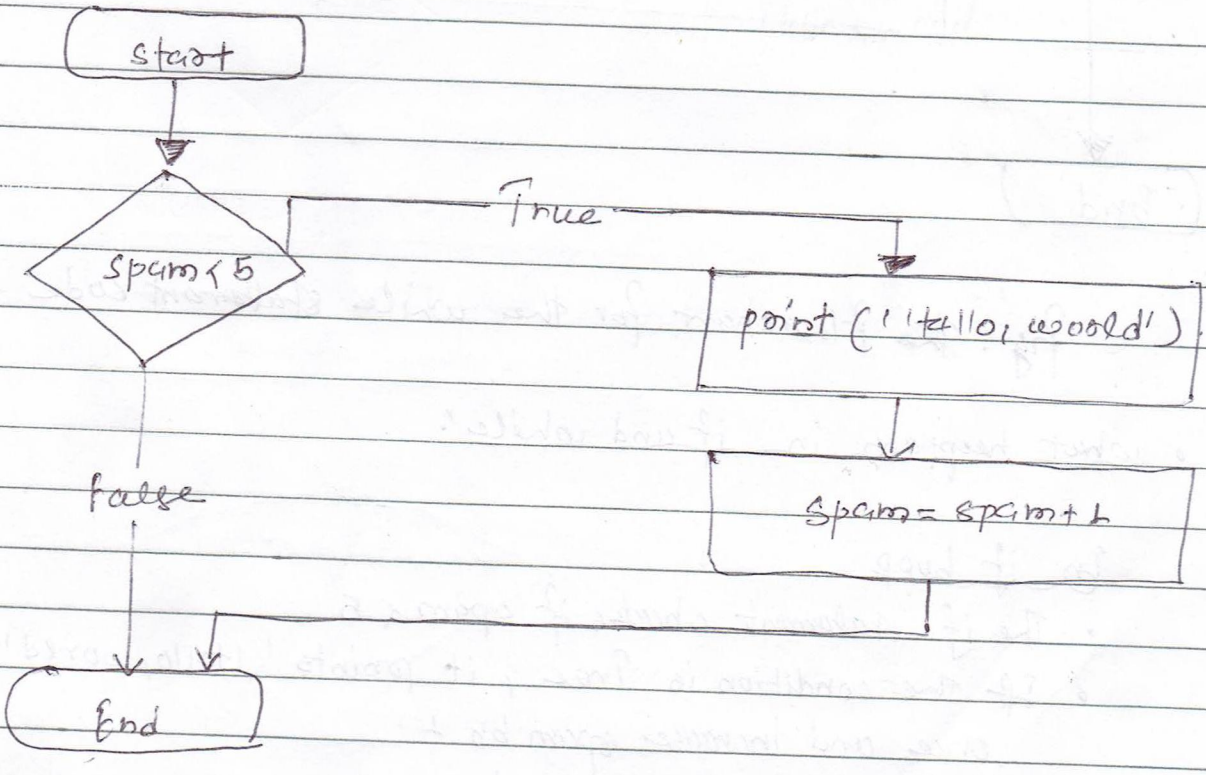


Fig: The flowchart for the if statement code.

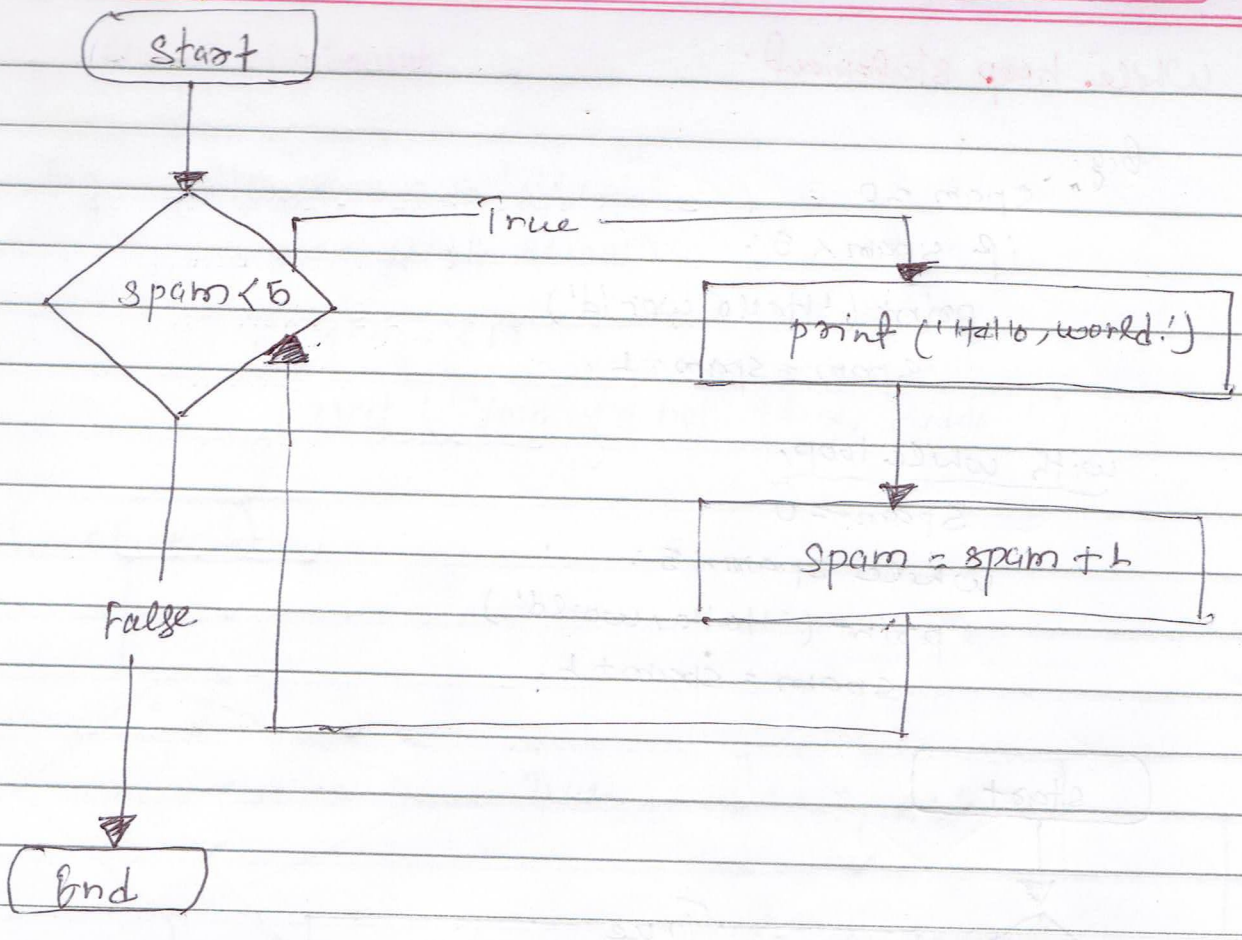


Fig: The flowchart for the while statement code.

- what happens in if and while:

In if loop

- The if statement checks if spam < 5.
- If the condition is True, it prints 'Hello, world' once and increase spam by 1.
- After that the program ends.

In while loop

- The while loop checks if spam < 5.
- if the condition is True, it prints 'Helloword' and increase spam by 1.
- The loop repeats as long as the condition is True.
- Once spam reaches 5, the condition becomes False and the loop stops.

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if vs while

Runs the code once if the condition is true

Repeats the code multiple times as long as the condition is true.

output

if

Hello, world

~~Hello, world~~

while

Hello, world

Hello, world

" "

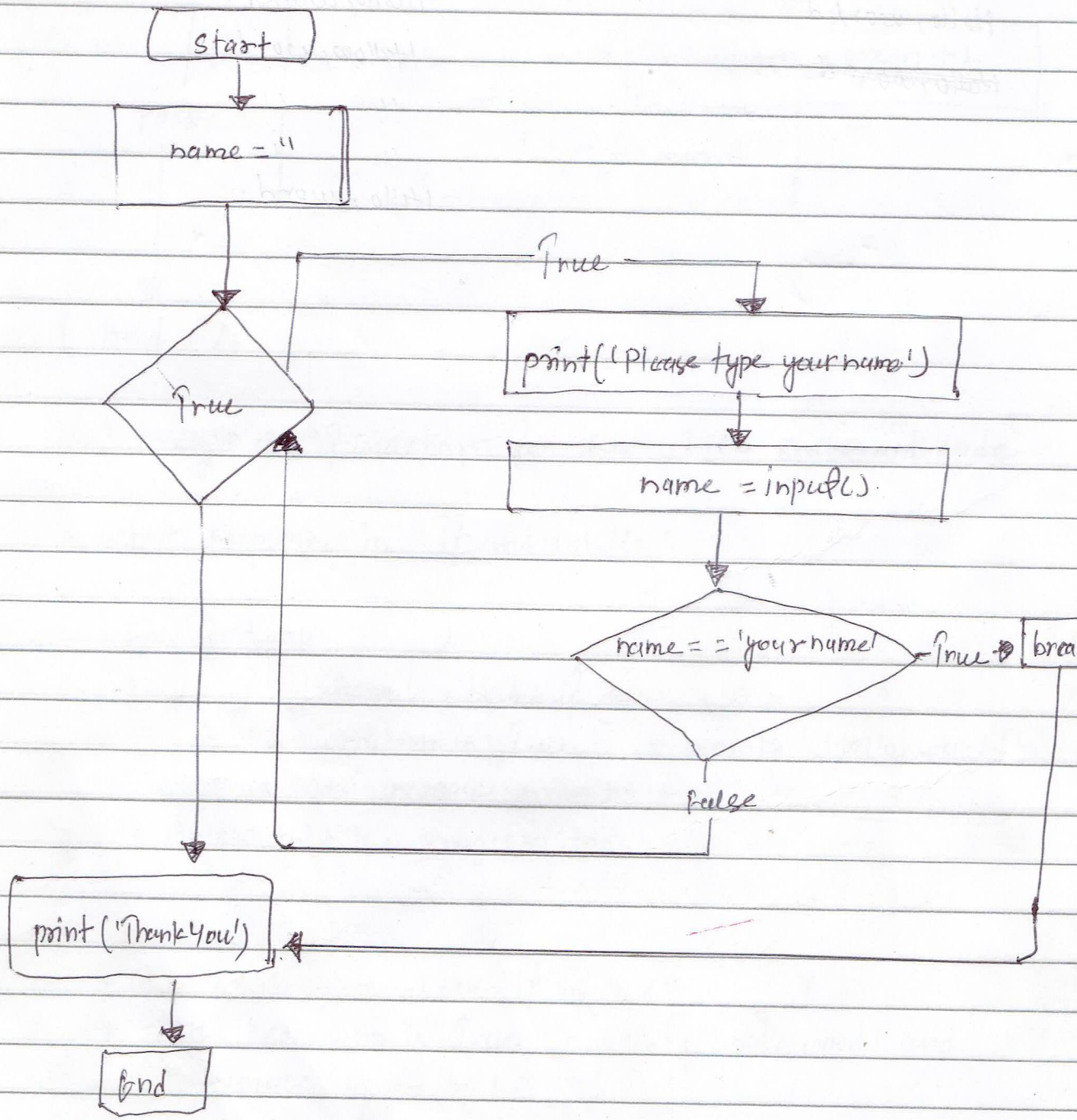
" "

Hello, world

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Break Statements

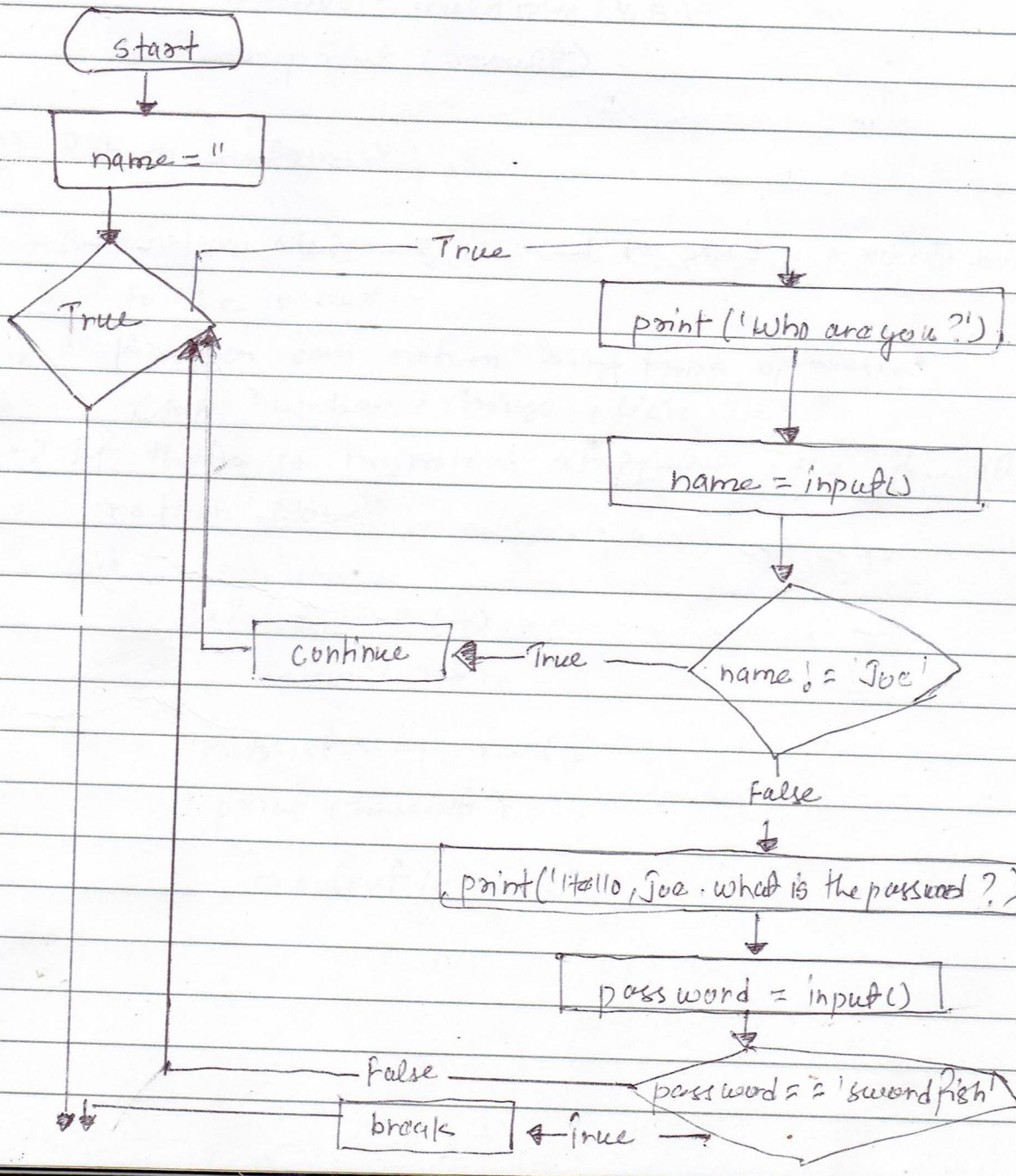
```
while True:
print('Please type your name.')
    name = input()
    if name == 'your name':
     break
    print('Thank you!')
```

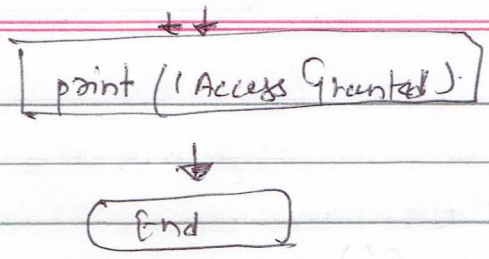


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Continue statement.

```
while True:
    print('Who are you?')
    name = input()
    if name != 'Joe'
    Continue print('Hello, Joe. What is the password?')
      password = input()
    if password == 'sword fish'
      break
      print('Access granted!')
```




print ("Access Granted")
End